Introducing computer vision for the humanities

The case of cross-written letters and the virtue of ‘going local’ with LLMs

Handout for the lecture at Linnaeus University, Tuesday 3 November 2026

Christoph Nicolai · mail@christoph-nicolai.de

Without any manual labelling and without a commercial cloud service, the cross-written letters can be separated and transcribed. That is what the lecture showed, in four parts: a computer sees an image as a grid of numbers and can answer questions about that grid. It learns the answers from pairs of examples (‘ground truth’), adjusting millions of numbers a little at a time. Where no labelled pairs exist, they can be built synthetically, and a multimodal model can read the result on one ordinary computer. The case study is my MA thesis, *Reading the Crossed Page* (Linnaeus University, 2026).

Contents

What is a digital image? 2

What do we want to know about an image? 3

What is a filter? 4

What is supervised learning? 6

Where do the pairs come from? 9

Separation done, now transcription 11

Time to add the humanities to the digital 15

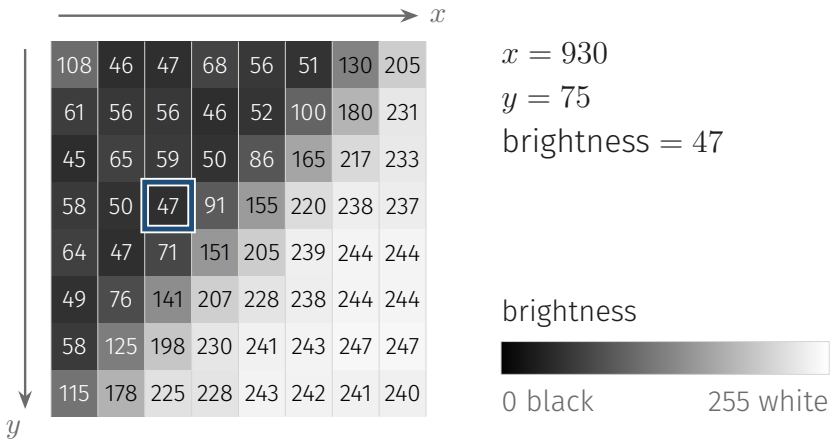
Glossary 19

Contact and licence 20

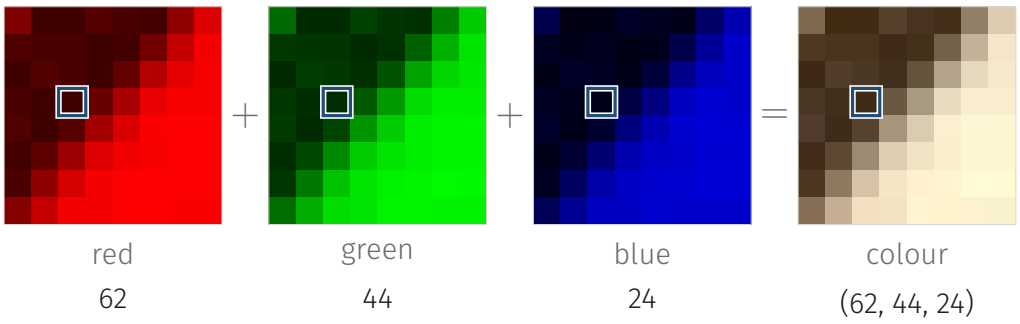
This handout and the workshop material:
github.com/cnicolai/cv-humanities-lnu-2026

What is a digital image?

A grid of pixels



- A scanned page is rows and columns of pixels. Each has an x and a y, counted from the top-left corner, and one number for its brightness: 0 is black, 255 white.
- The pixel at $x = 930$, $y = 75$ has brightness 47.
- Colour is three numbers per pixel, red, green and blue. This pixel's 62, 44 and 24 make a dark brown; as grey they become 47.
- Every question about an image is a question about these numbers.



The '19' of the date line, George Stephens to Joseph, 19 November 1868; Huseby Archives, Linnaeus University, E I:3a. Grey from colour: $0.299\,R + 0.587\,G + 0.114\,B$, weights that follow the eye, which sees green brightest and blue darkest (BT.601).

What do we want to know about an image?

Classification, detection, segmentation

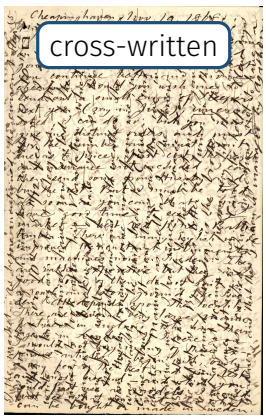
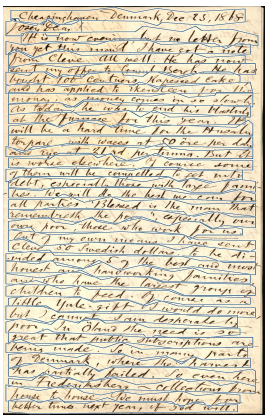


Image classification

one label
for the whole image



Object detection

a box or polygon
around each object



Semantic segmentation

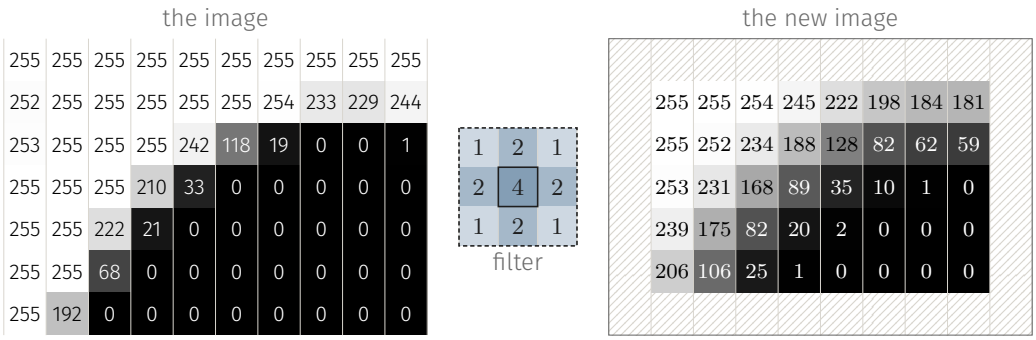
a class
for every pixel

- **Classification:** one label for the whole image. Cross-written or not? A map, a newspaper, a book page?
- **Detection:** a box or polygon around each thing, here around each line of the page, found by Riksarkivet's line detector.
- **Semantic segmentation:** a class for every pixel: first layer, second layer, both, or paper.
- All three questions can be put to the same page; which one you ask depends on what you want to know.
- The HTR community calls line detection 'segmentation'. Here the word keeps its strict meaning: a class for every pixel.

Huseby Archives, Linnaeus University, E I:3a, 19 November and 23 December 1868; line polygons: Riksarkivet's YOLO detector (htrflow); layers: the thesis model.

What is a filter?

A filter sliding over the image



- A filter is a small grid of numbers, here nine. Laid over nine pixels, each pixel is multiplied by the number above it and the products are added: one number for the pixel under the centre.
- Divided by 16, the sum of the nine, that is a weighted average of the neighbourhood: a blur. The rim of the image gets no value.
- Slide the filter over every position and the new image is complete. Whatever the numbers, the operation is a convolution.
- A 1 in the middle and zeros around it change nothing. A 5×5 filter averages wider. Moving the filter two pixels at a time, along and down, gives a new image of half the width and half the height: that is how a network shrinks its images, layer by layer.

A window of 10×7 pixels; integer weights, the sum over 16 rounded.

The two edge filters on the cathedral



$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & \boxed{0} & 2 \\ -1 & 0 & 1 \end{bmatrix}$$



$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & \boxed{0} & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

vertical edges horizontal edges

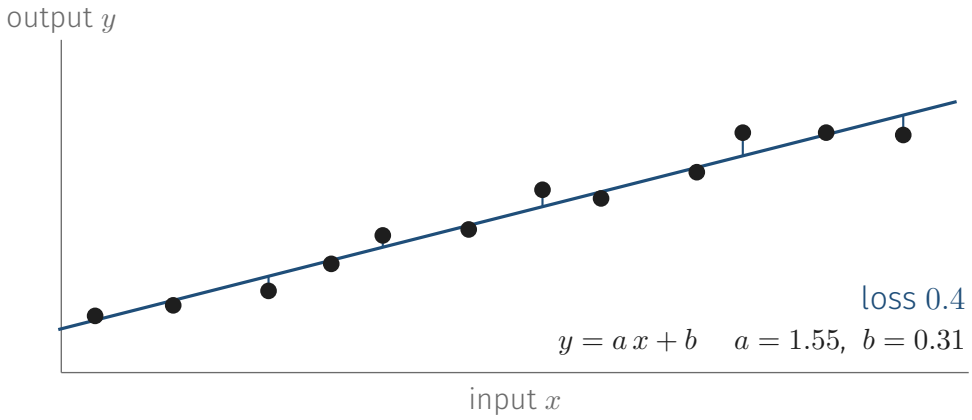
nine numbers each, chosen by our understanding of an edge

- An edge is an abrupt change of brightness, and these filters detect it. Minus on the left, zero in the middle, plus on the right: the filter compares the right of its window with the left. On plain sky the sum is near zero; where sky meets brick it is large.
- The size of each sum, drawn as grey, inverted so that it reads like ink on paper: the darker the pixel, the stronger the edge. The vertical edges stand out: the sides of the tower, the spires, the pillars of the window. The cornice, which runs left to right, vanishes.
- Turn the nine numbers by ninety degrees and the horizontal edges appear instead. A slanting edge, like the spires, changes brightness both ways and shows in both maps.
- These are the Sobel filters: nine numbers chosen by a person from an idea of what an edge is.

The top of the cathedral tower at Stortorget, Växjö. Photo: Klugschnacker, 2009, CC BY-SA 3.0, modified (cropped, grey, filtered).

What is supervised learning?

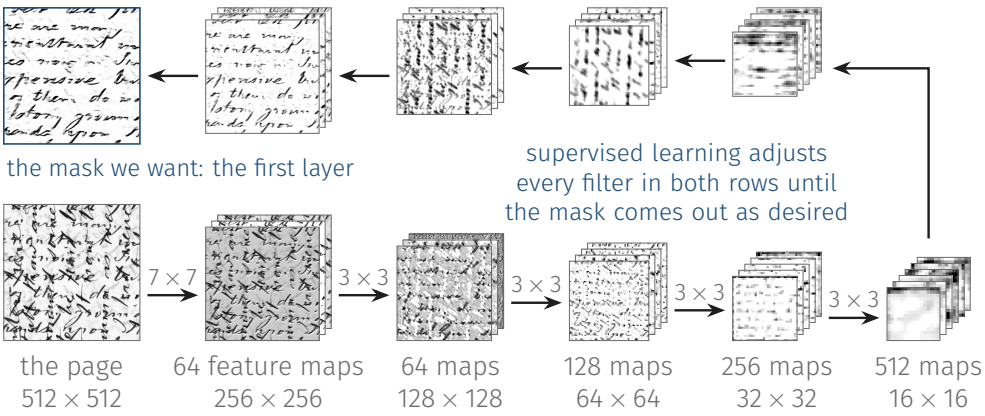
How the pairs set the parameters



- Often we write a mapping ourselves, as with the edge filter. What would the rule be for ‘the ink of the second layer’? Supervised learning turns it round: pairs of an input and its desired output, the labels, define the mapping.
- The simplest case: a line $y = ax + b$, two adjustable parameters, twelve points. The gaps, squared and added, are the loss: 32.8 at the first guess.
- Change a and b iteratively so that the loss gets smaller: after a few steps it is 0.37 and the line sits among the points, at $a = 1.55$, $b = 0.31$.
- The step size is the learning rate, the method gradient descent. The computer never sees the whole landscape of the loss, only which way is downhill where it stands; with millions of parameters there are many valleys, and the start matters.

Twelve invented points; gradient descent on the slope and the intercept.

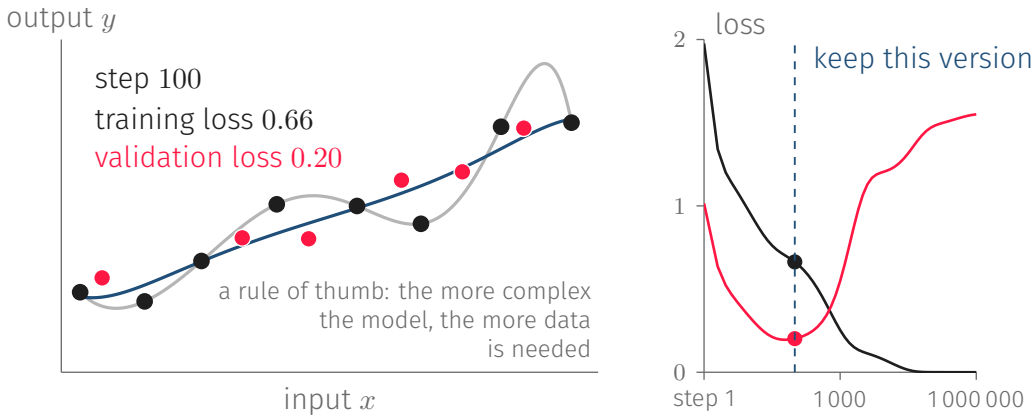
Supervised learning tunes all of it



- For images the mapping is a convolutional network: filters as on the cathedral, but the examples set their numbers.
- The encoder stacks layer after layer of filters. Their outputs, the feature maps, halve in size and double in number, down to 512 maps of 16×16.
- A head turns the maps into the answer's shape: a class, boxes or, as here, a mask the size of the page. Encoder and decoder together: a U-Net.
- The same loop as the line tunes every filter: 24.5 million parameters in my thesis model, each changed a little at every step.

ResNet34 encoder (He et al., 2016); U-Net (Ronneberger et al., 2015). Feature maps of my thesis model on a tile of the 19 November 1868 page; Huseby Archives, Linnaeus University, E I:3a.

Stop where the validation loss is lowest

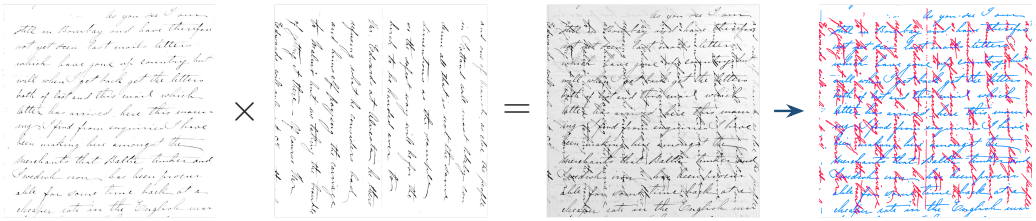


- Eight examples and a model with eight parameters, as many as there are examples; trained step by step, the training loss falls and falls, until the curve passes through every point.
- Six examples held back, from the same source, are never learned from, only measured: the validation loss. It falls at first, is lowest after 100 steps, then rises as the model learns the eight points by heart. That is overfitting.
- The validation loss says when to stop and which version to keep: the gentle curve of step 100, not the perfect one. A rule of thumb: the more complex the model, the more data is needed.
- So the labelled examples are split before anything starts: training, validation, and a test set opened once, at the end, for the number you report.

Eight invented points and six more from the same source; the model is a curve with eight adjustable parameters.

Where do the pairs come from?

Make the inputs from the outputs



a plain page
Joseph, 14 Nov 1868

another page in the same hand, turned by 90°

then aged: simulating a real paper background

the answer, for every pixel

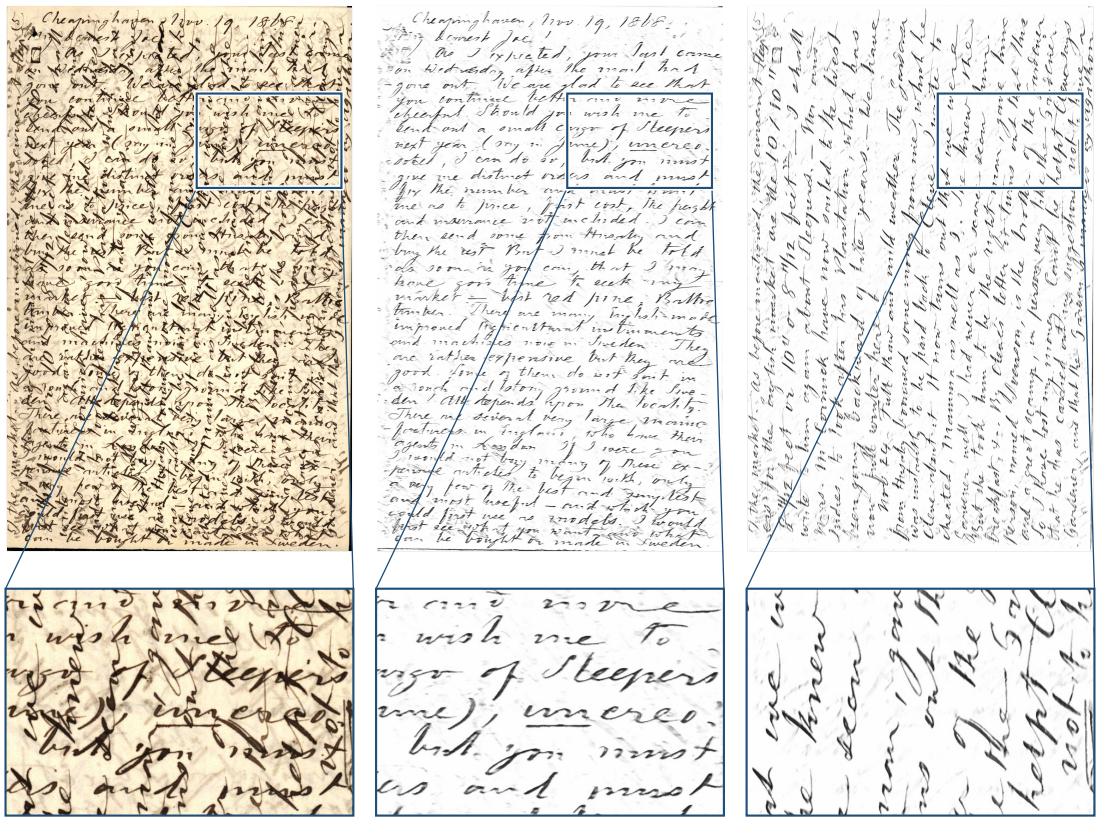
- One pair for this task is a crossed page and, for every pixel, which layer it belongs to. Creating this by hand would be extremely laboursome.
- The reverse mapping is relatively easy: turn a plain page and lay it over another. So the answer is made first and the input built from it: synthetic data.

1. Take a plain page and another in the same hand, turned by ninety degrees.
2. Overlay the two images. Where ink meets ink the point darkens twice.
3. Age the result: fade the ink unevenly by up to half, add bleed-through, foxing and a paper tone.
4. Keep the mask that comes with it: first layer, second layer, both, paper.

- From 208 plain pages in Joseph's and George's own hands: 1 200 crossed pages, 1 020 to train and 180 to validate.

The pages in Joseph Stephens's hand; Huseby Archives, Linnaeus University, E I:2.

One page, two layers



George to Joseph, 19 November 1868 — the horizontal layer — the vertical layer

- The page of 19 November 1868 and the two layers my model found: the horizontal writing, then the vertical one, from the page turned by ninety degrees and run through the same network.
- Most of a page is paper, so a mask that marks nothing can be almost right. The loss is built against that trivial solution: missed ink costs more than kept paper gains.

Huseby Archives, Linnaeus University, E I:3a, 19 November 1868; layers: the thesis model.

Separation done, now transcription

In the cloud, or on your machine: Gemma 4

- Two ways to read a layer: a line detector and a line reader such as TrOCR, one line at a time; or a multimodal language model, which takes the whole layer image and writes the text.
- Such a model runs in a company's data centre or, if it is open, on your own computer. Mine: Gemma 4, 31 billion parameters, in LM Studio on one Mac, forty seconds a page.

A decision.

- **data protection:** where the scan goes
- **ethics:** intimate letters in a company's data centre
- **reproducibility:** the same answer next year
- **independence:** a service can change or disappear

Gemma 4, 31B, 8-bit quantised, temperature 0 (Google, 2026); LM Studio.

Neither fine-tuned: Gemma errs far less

$$\mathrm{CER} = \frac{S + D + I}{N}$$

S substitutions, D deletions, I insertions
N characters in the text as written



(3 + 1 + 1) / 24 = 21 %

- The character error rate (CER): the single-character edits that turn the model's text into the text as written, divided by the characters written.
- Seven test pages, fourteen layers, 333 lines; both engines used as published, neither tuned to these hands.

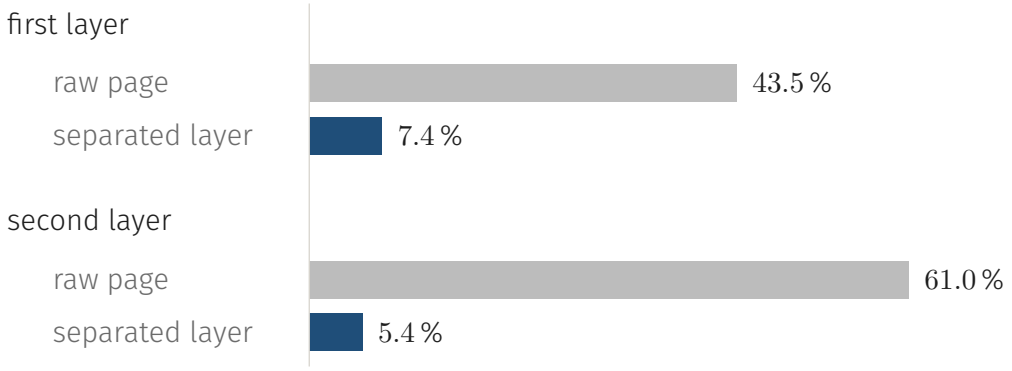
Character error rate per test page, in per cent

	first layer			second layer		
	raw page Gemma	layer Gemma	layer TrOCR	raw page Gemma	layer Gemma	layer TrOCR
George, 10 May 1868	26.5	14.3	38.6	84.5	6.5	47.0
George, 21 July 1868	74.7	11.1	41.3	66.6	10.9	38.1
George, 31 October 1868	31.0	5.7	27.2	99.8	4.2	30.3
George, 1 December 1868	74.3	8.9	45.7	65.5	3.4	26.5
Joseph, 6 June 1868	46.2	3.8	23.7	19.4	4.2	23.8
Joseph, 16 October 1868	35.0	4.1	28.4	15.5	3.8	26.1
Joseph, 3 December 1868	16.5	4.2	18.7	75.6	4.8	31.0
mean of the seven pages	43.5	7.4	31.9	61.0	5.4	31.8

'raw page': the crossed page read without separation. The lecture's 7 and 33 per cent weight every character alike instead of every page. The example line above: George Stephens to Joseph, 10 May 1868, first layer; Huseby Archives, Linnaeus University, E I:3a.

- Only seven pages, and the same pages chose which version of the separation model to keep: a first result, not a benchmark.

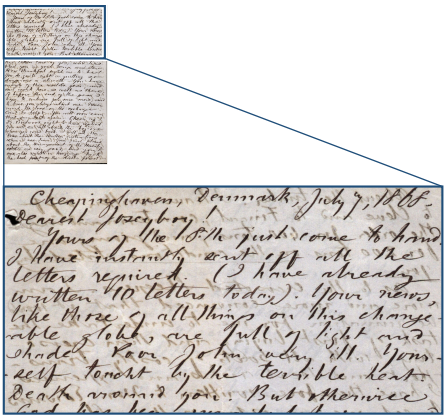
Seven pages: from a half to a twentieth



- The same reader on the raw crossed page and on the separated layer: 44 and 61 per cent of the characters wrong, against 7 and 5.
- The separation made the page readable.

Means over the seven test pages; character error rate.

Fluent text that is not on the page



Copenhagen, Denmark, July 7, **1885**.
Dearest **Jocelyn**,
Yours of the **15th** just come to hand.
I have instantly sent off all the
letters **expired**. (I have already
written 10 letters today). Your news,
like those of all things on this change-
able globe, are full of light and
shade. Poor John **was** ill. Your-
self **tonight** by the terrible heat.
Death **hovered near** you. But otherwise

- The model's reading of a plain page, 7 July 1868: Copenhagen for Cheapinghaven, 1885 for 1868, Jocelyn for Joseyboy, 'death hovered near you'. Fluent, plausible, wrong, and nothing marks it.
- A meaningful invention is harder to find than a garbled string; this is a problem of reading with large language models. A rule to mark illegible passages helps; a person still checks every page.
- Where the method stops: a second layer in red ink over black stayed unreadable. Blue paper with low contrast did not matter.

George Stephens to Joseph, 7 July 1868; Huseby Archives, Linnaeus University, E I:3a.

Time to add the humanities to the digital

Cross-writing, in their words

'Well, Hetty, now I think you will be put to it to make out all that chequer-work'

Jane Austen, *Emma*, 1816

'A lover -- a truly happy, ardent passionate lover -- can stand crossing and scented note-paper and both are trials to the male heart.'

Saturday Review, 1861

'Whatever you do, don't cross! Remember the old proverb "Cross-writing makes cross reading"'

Lewis Carroll, *Eight or Nine Wise Words about Letter-Writing*, 1890

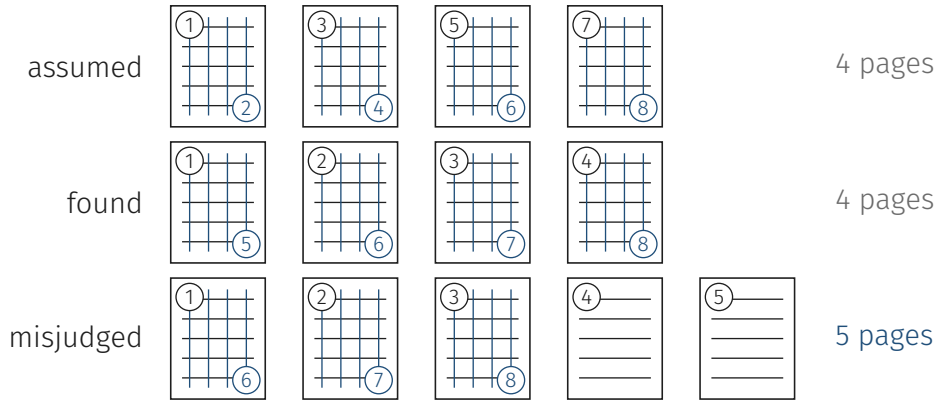
'Even cross-writing … could be endured for the sake of love.'

Maggie Kalenak, *Journal of Victorian Culture*, 2023

- George Stephens, professor in Copenhagen, and his son Joseph, building railways in India from 1860: 112 letters, 531 pages, 310 of them crossed, by both hands.
- Some letters cross only on the last page, where the paper ran out; others from the first line: an expedient, or a plan.
- The usual explanation is paper and postage: crossing doubles the text on a sheet and saved under the tariffs before and after 1840. After the reform it was called poor taste, and went on.

Austen, *Emma*, vol. II, ch. 1; *The Saturday Review*, 23 February 1861, as quoted in Kalenak 2023, p. 252; Carroll's rule nine.

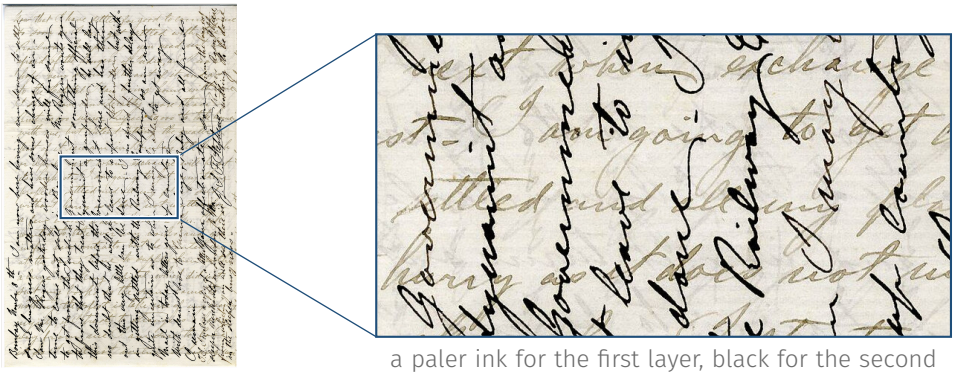
Found: all horizontal layers first



- I had assumed that a writer filled a page, turned it and carried on. Once the layers were apart, some letters showed another order: every horizontal page first, then back to the first page to cross.
- That means guessing how many pages to write before crossing. One letter guessed one too many: five pages for eight layers, where four would do.
- Had saving been the motive, each page would have been filled before the next. This is only true where a letter spans several loose sheets, and drying time is another explanation.

Huseby Archives, Linnaeus University, E I:2 and E I:3a.

Two inks on one page: a gesture of care



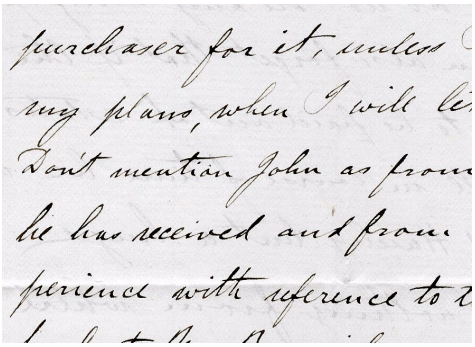
Joseph to George, 5 November 1868

a paler ink for the first layer, black for the second

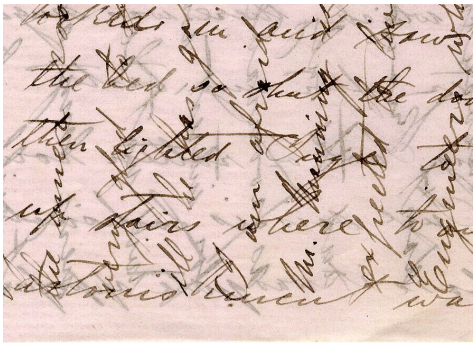
- A paler ink for the first layer, black for the second, in some crossed letters, not most: the writer accommodating the reader in the material itself.
- Keeping a crossing readable was a skill. Joseph's fine nib and even spacing make a calm grid in which each writing can be followed; George's pages shift in density until the letter shapes vanish.

Huseby Archives, Linnaeus University, E I:2, 5 November 1868.

Long t-bars guide the eye



a plain page, **28 November 1867**



a crossed page, **26 December 1867**

longer, it seems: an impression, not a measurement

- Joseph's t-bars run far beyond the letter on a plain page already. On a crossed page they lead the eye along the line and set the two writings apart.
- They seem longer on the crossed page, written four weeks later in the same hand: an impression, not a measurement.
- The thesis's closing thought: intimacy may be not only a licence for crossed-writing but also a reason for it. For a father and son who met once in ten years, the page was the only space they shared.

Huseby Archives, Linnaeus University, E I:2, 28 November and 26 December 1867.

Glossary

Pixel
One cell of the image grid, with its brightness, or its three colour values, as numbers.

Filter, convolution
A small grid of numbers slid over the image; at each position the products with the pixels beneath are added up. The operation is a convolution.

Feature map
The image a filter produces. Each layer of a network makes many.

Supervised learning
Learning a mapping from pairs of an input and its desired output.

Label, ground truth
The desired output of an example, provided by a person or by construction.

Parameter
A number inside the mapping that learning adjusts. The line had two, the thesis model 24.5 million.

Loss
One number for how wrong the mapping is on the examples. Training makes it smaller.

Gradient descent, learning rate
Adjusting the parameters a little downhill, step after step. The learning rate is the size of a step.

Overfitting
Learning the training examples by heart, so that new examples are answered worse.

Training, validation and test sets
The examples learned from; those held back to decide when to stop; those opened once, at the end, for the reported result.

Encoder, head, decoder
The layers that turn an image into feature maps; the layers that turn the maps into an answer; a head that builds an image back up is called a decoder.

Semantic segmentation
A class for every pixel. Detection draws a box around each thing; classification labels the whole image.

Character error rate
The single-character edits needed to correct a transcription, divided by the characters in the text as written.

Multimodal model
A language model that also takes images: it reads a whole page and writes the text.

Contact and licence

– Christoph Nicolai · mail@christoph-nicolai.de

– The case study is my MA thesis *Reading the Crossed Page* (Linnaeus University, 2026).

– This handout and the workshop material: github.com/cnicolai/cv-humanities-lnu-2026

– Text and diagrams: CC BY 4.0, Christoph Nicolai 2026. Photographs excepted: the letter scans, Huseby Archives, Linnaeus University (public domain; not offered as a dataset); Stortorget, Växjö: Klugschnacker 2009, CC BY-SA 3.0, modified.